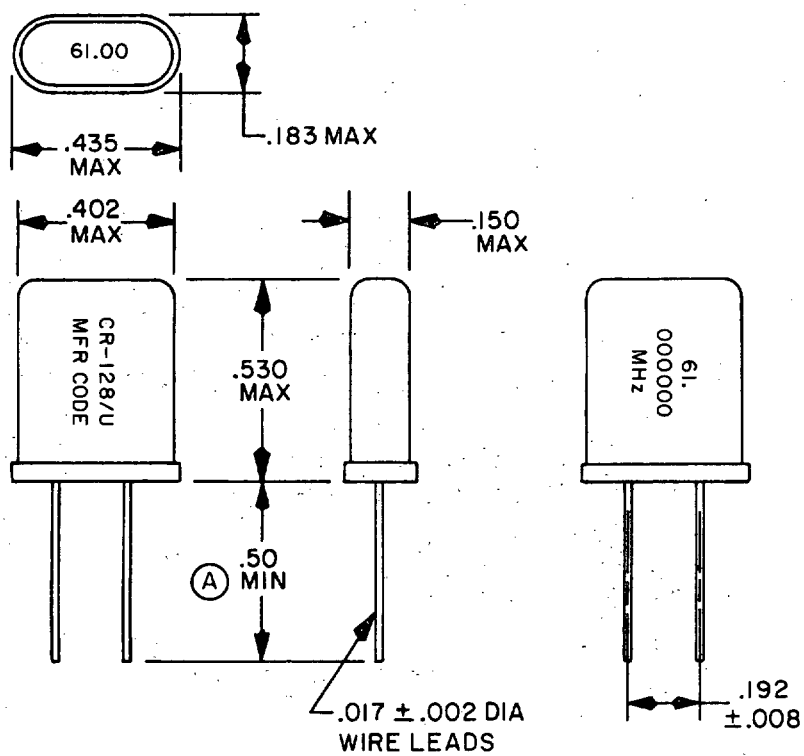
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-128/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 17 to 61 MHz; fundamental; noncontrolled; series resonance.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-128/U.

Ⓐ denotes changes

FSC 5955

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Ⓐ TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure 1.
Frequency range	17 to 61 MHz, inclusive
Capacitance, shunt	7 pF, maximum
Mode of oscillation	Third mechanical overtone
Government reference standard test set	TS-683/TSM
Rated drive level	2.0 ±0.4 mW
Calibration values:	
Resistance	40 ohms
Resistance voltage drop	0.28 volt
Resonance	Series
Operating temperature range (noncontrolled)	-55°C to +105°C, inclusive
Operable temperature range	-105°C to -125°C, inclusive
Frequency tolerance (operating temperature range)	±50 ppm
Equivalent resistance	40 ohms, maximum
Unwanted modes	Method II
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent, or 2 ohms, whichever is greater
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent, or 2 ohms, whichever is greater
Thermal shock	±50 ppm
Aging:	
Frequency change permitted	±10 ppm

1/ Method 204 of MIL-STD-202, test condition A, 0.015 inch (0.03 inch total excursion) 10 to 55 Hz; or an amplitude to produce 10 gravity units, 55 to 500 Hz, whichever is the limiting value.

NOTE: These crystal units are for use in the AN/ARN-52 equipment.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 85
DLA - ES

User activities:
Air Force - 17, 19, 84

Agent:
DLA - ES

Preparing activity:
Army - ER

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